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Development of Attachable Modules for Robotizing Daily Items
-Person Following Shopping Cart Robot-

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Abstract **- We introduces a development of attachable module for robotizing daily items. We aim at realization of robotizing daily items aiming at popularization in a robot's everyday life. We set the shopping cart as the concrete target. The aim of this study is to develop a porting system for large-sized facilities, where the object item is autonomously carried by a person following ("after you") porting robot. The driving module and the sensor module are developed for development of a shopping cart robot. The outdoor running of the robot is attained by reforming a drive mechanism. Manual control is needed for the risk aversion at the time of a reckless run, or an operation test. The driving mechanism is used the compatible system which can change manual control and automatic control. This sensor has stereo camera and is used for person following system. The following system is the method of using together distance information and the information on a color. Template matching using the HSV color system as information on a color is performed. We do the operation test about the driving mechanism, the compatible control system, and the person following operation, and performed the check of the robot of movement. By the experiment, while the shopping cart robot carried baggage, we check that it was possible to perform following operation.**

Index Terms –Robotizing, shopping cart, modules, person following

I. INTRODUCTION

A robot is recognized in everyday life and many robots aimed at using it in daily life are studied. Many robots have moving functions or information processing functions in own case. One robot has all the required components and functions. A life of people will become more convenient because many robots are popularized into daily life. However, room becomes narrow instead of the number of robots increasing. People will feel discomfort to robots because robots jostle in space. Even if that is not right, in order to be popularized in daily life, the robot must be a design which does not give a discomfort to people. In order to develop comfortable design of case in daily life, many labors and costs are needed. Finally, only robot's functions should be popularized in daily life. In the other words, the transparent robot which is not visible makes neither narrowing space nor a man feel a discomfort. However, development of a transparent robot is impossible.

Therefore, in this research, we propose robotizing daily items. The daily items already used in everyday life are designed comfortable, and are regularly used by people. If the daily items become a robot as it is, it becomes without taking a place and people can also contact a robot comfortable. What is necessary is for a transparent robot not to carry a chair, but for the chair itself to be robotized, and to be able to run by itself. We develop modules based on daily items (i.e. desk or chair). The developed driving or sensor modules are attached to daily items. We robotize daily items by attaching only robot's functions. We aimed to reduce a labor and cost for development robot's private case.

The conception picture of robotizing daily items is shown in Fig.1. In Fig. 1, sensors, such as two cameras, have detected the target person in the robotized room. Based on the information from sensors, the robotized daily items around a target person perform each work. The wagon robot carries books for target, and shopping cart robot follows target during carrying baggage. Chair robots are aligned so that it may not become a target's obstacle. These robots are attached driving or sensor modules and robotized. It means robotizing daily items.

Ohara et al. [1] also use the same method of attaching a driving module to the existing thing. They add a conveyance function to existing thing. They are developing the module based on the information from a server.

However, we attach sensor module to daily items using stereo camera. We aim at robotizing daily items which can follow target person during carrying baggage without information from a server. In order to realize robotizing daily items,

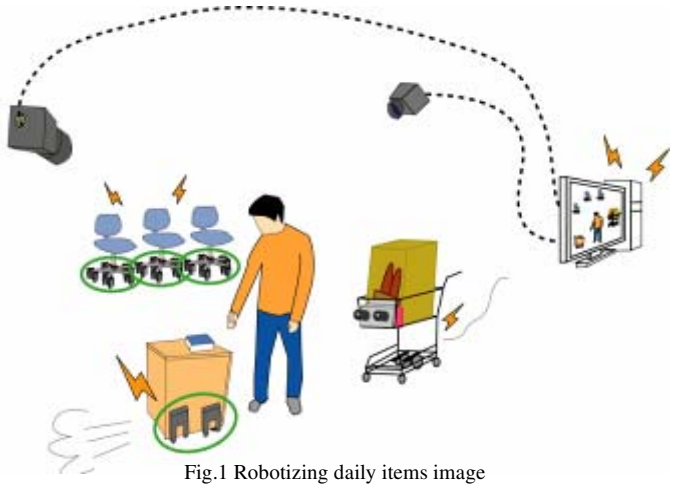


Fig.1 Robotizing daily items image

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shopping cart is used as a concrete subject matter. We aim at development of the system which the shopping cart robot can follow target person during carrying baggage in supermarket or airport, etc. We have developed the precedence experimental model of the driving module which based on the shopping cart [2]. We apply the person following system which Ito et al. [3] studied as a sensor module function.

II. Construction of a Shipping Cart Robot

We show the composition of the shopping cart robot under development in Fig.2. On the occasion of development of a shopping cart robot, we focus attention on the driving function and develop the driving module. The driving module is shown the portion expanded to the lower left of Fig.2. Since the shopping cart is used as the base, the standard of maximum loading capacity is 50kg. Since it is a person following system, we aim target speed at 4km/h which is people's walk speed. Moreover, we consider that the robot runs the outdoor. The structure of the axle part is developed which can allow the robot to run stably in the irregular ground or a slope.

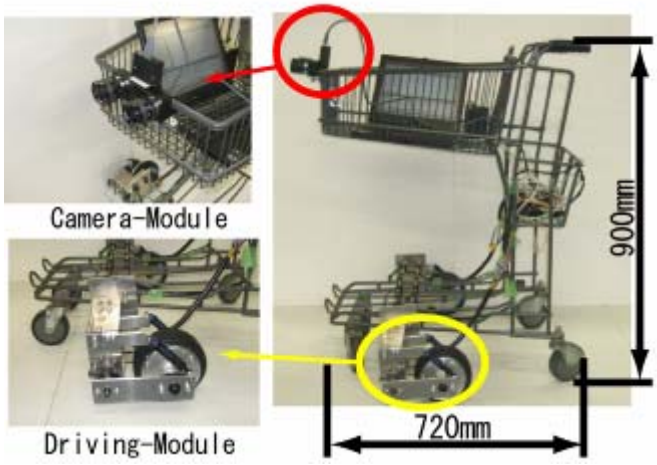


Fig.2 Prototype robot & modules

Table 1 specifications of the robot

Laptop PC	
OS	Windows XP
CPU	Pentium M 1.6GHz
Memory	1.5GB
Driving module	
Maximum speed	1m/s
Motor	TAMIYA, INC. 540K75
Stereo camera	
Company	Videre Design
Product name	MEGA-D
Interface	IEEE 1394
Frame rate	26.5fps
Field angle	50deg
Image size	QVGA(320*240)

In this research, an autonomous run is controlled by a laptop PC. However, manual control may be required because of the run control at the time of a reckless run, and the quality assessment of the driving module. We divert the system of the existing manual control. The robot is controlled by the change type of laptop control and operation by person.

The upper left of Fig.2 is a stereo camera used as the sensor module. In this research, a person following system is constituted by an image processing from a stereo camera. Therefore, the stereo camera is attached front of the cart. The distance and position to a following target person are calculated by image processing based on disparity information by binocular parallax and color information by color image. The stereo camera which processes the image size of QVGA (320x240), and can acquire a disparity image by 26.5 (fps). The field angle is a 50-degree lens. The stereo camera is connected with the computer via an IEEE1394 cable. The camera power is supplied from the battery.

The specifications of the robot are shown in Table1. All the control is performed by only one laptop PC. Below, we describe the driving module, the control system, and the person following system, respectively.

III. DEVELOPMENT OF DIRIVING MODULE

In Fig.3, the driving modules are shown. The design 1 of the driving modules reported by [2] has a tire driving system. In this system, a motor is directly in contact with a tire. The tire driving system is shown in Fig.4. This system has advantages of the high torque and easy power transfer. The weight which can be loaded is set up with 50kg as a driving function. We use a motor made from TAMIYA, INC. which is low cost and produce the high torque. Since an encoder is not mounted in the motor, an encoder is set and measured the number of rotations of the motor. However, there are some problems. One of problems is sand between a tire and a motor in outdoor running. Therefore, the driving module is a idling of the tire and the motor for sand. Then, the design 2 of the driving module is constituted by a motor, a tire, a gear, and a belt. A driving power is transferred by the tire using a gear and a belt. If the tire of design 2 has sand, the robot can run outdoors for the design 2. Although the full length of the design 1 is 320mm, the design 2 is miniaturized and set to 250mm.

Another problem, in design 1, the driving module is attached inside of the cart because we don't want to change the cart size. However, by attaching inside, the ease of attachment and detachment has been lost. And the robot's turning performance worsened. Based on the experience with a precedence experimental model, a joint is designed with easy attachment and detachment. The droving module is connected to the outside of a cart.

The running test of the developed driving module is done. In order to aim at the operation test of the driving module, the manual control is used for the run. The experimental situation is shown in Fig. 5. In the section of 1 to 4 of the figure, the robot runs an indoor hall. Then in the section of 5 to 8, the robot runs the outdoors as it is through a slope. This is the

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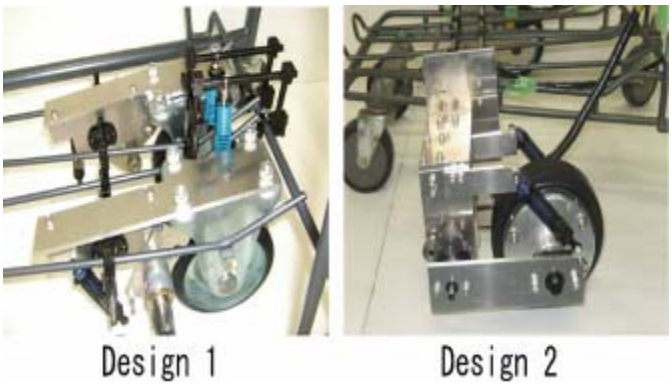


Fig.3 Driving modules


Fig.4 Tire driving system

examination to which the shopping cart robot runs not only indoors but the outdoors. In this figure, we show the robot is able to run regardless of indoor and outdoors by attaching this driving module.

IV. COMPATIBLE CONTROL SYSTEM

In run control of the prototype, in order to perform person following automatically, it is needed that the robot can move autonomous. In this research, the system of the existing manual operation is diverted for the run control at the time of a reckless run, and the quality assessment of the driving module. We use the compatible control system in which the change to laptop PC control and operation by person is possible.

The design figure for the compatible control system is shown in Fig.6. For the urgent avoidance and an emergency halt at the time of a reckless run, a manual control of a radio type is used. By controlling by a radio type, even if the robot will be in a reckless run state when an operator is far from the robot, he can control the robot and perform an emergency halt. Moreover, a battery is prepared separately for the laptop and the motor for running. Even if the battery of a laptop passes over hours of operation and forced to terminate, an operator can manual control by using motor's battery and make the robot run to the charge point.



Fig.5 Running examination for the driving module

In order to realize compatiblization, the motor driver is diverted used by automatic control for manual control and take as a common motor driver. In Fig.6, in order to give priority to the signal of manual control when a signal is sent simultaneously, a relay circuit is used.

When a signal is sent using a relay circuit, a problem is occurred. There is a problem that the voltage of the signal of a receiver end used by manual control is lower than the voltage of the relay circuit of operation. Then, the operational amplifier is used between the receiver and the relay circuit, and the relay circuit can operate by making voltage amplify. In fact, although the voltage of the relay circuit of operation is 1V, the voltage in the receiver is 250mV. The magnification rate is set up 144 times using two operational amplifiers. Just before the rely circuit, voltage is amplified to 3.6V, and the relay circuit operate.

More over, the signal of the receiver is a PWM signal which is a cycle of 55Hz. Therefore, when the relay circuit is connected directly, a switch is change with the cycle of 55Hz. And it is impossible to give priority to manual control. Then, the PWM signal is smoothed by facing across integration circuit between two operational amplifiers. The relay circuit can operate stably.

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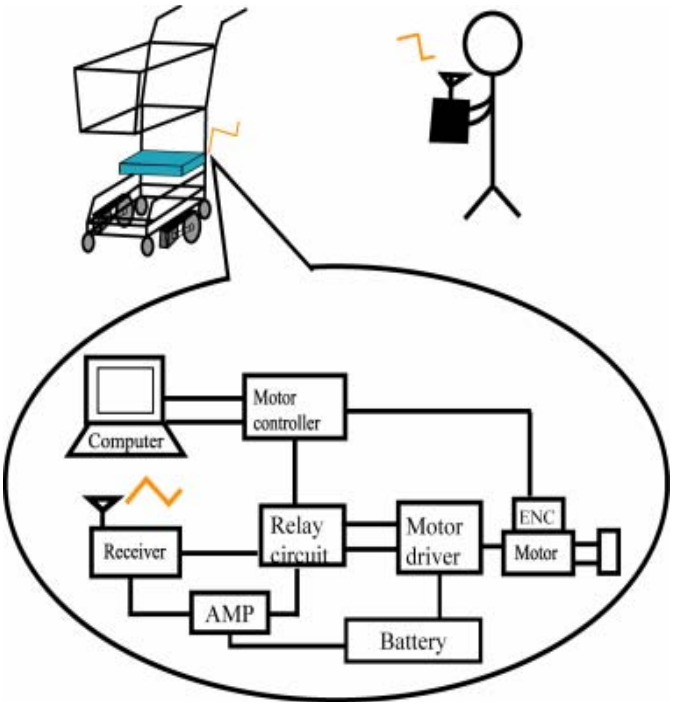


Fig.6 Compatible control system image

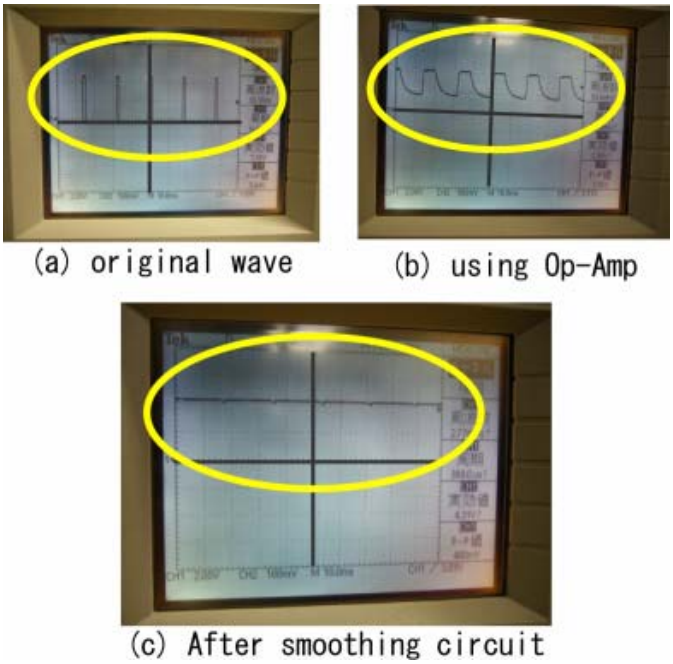


Fig.7 Original wave pattern & after using smoothing circuit pattern

The wave arranged patterns are shown in Fig.7. The image (a) is the original wave of the receiver. This cycle is 55Hz. The (b) shows the wave using op-amp. This wave is amplified to 3.6V. However, the frequency of the wave is still 55Hz. In image (c), after smoothing circuit, the wave is smoothed and the relay circuit is operated.

The running test of the built compatible control system is done. The sequential photographs at the time of putting baggage on the shopping cart robot and it running indoors are shown in Fig.8. The character of OFF and ON in the figure shows the state of the switch of the transmitter. The first photograph of Fig.8 is the run by laptop control, and the switch of manual control is turned OFF. In the second of the figure, when the robot becomes clashing with a wall, the operator turns ON the switch by the side of the transmitter, and stops the robot's run. Then, the operator controls the robot to run to him using manual control. By this examination, we are able to check the scram of the robot and urgent avoidance movement which used manual control.

V. PERSON FOLLOWING ALGORITHM

The method of recognizing a target person has many kinds, and is verified at various angles. Schulz [4] is performing person following based on the data of a distance sensor. By using a laser range finder, a target person's high range finding accuracy and high resolution can be obtained. However, this method will tend to generate occlusion, if person following is performed in a confused indoor environment. Person detection rate decreases extremely at the time of occlusion generating. However, in the environment where two or more persons exist, it becomes difficult to

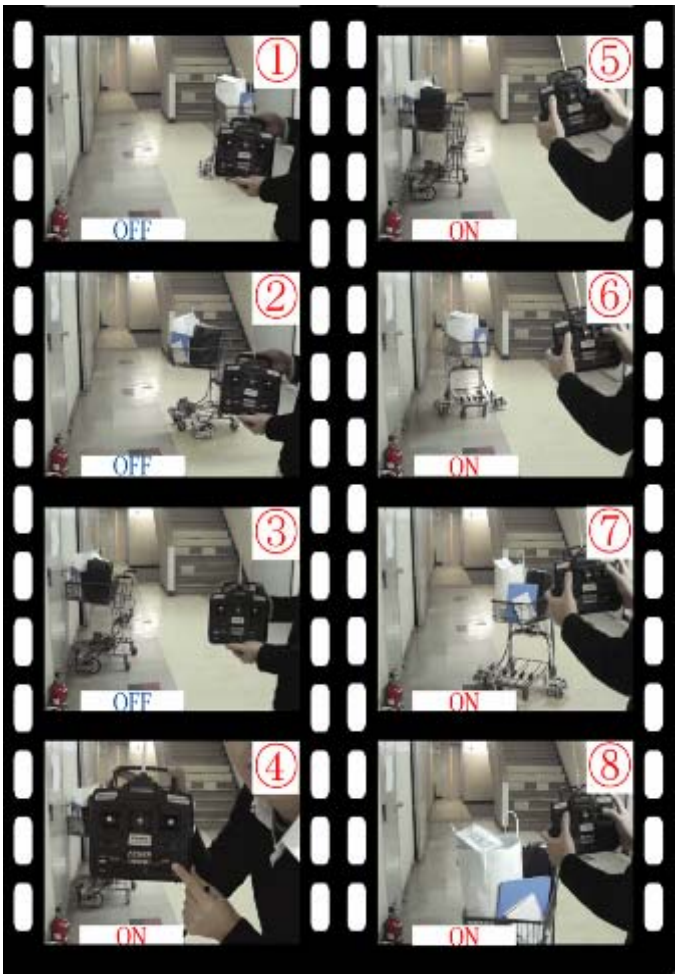


Fig.8 Examination of compatible system

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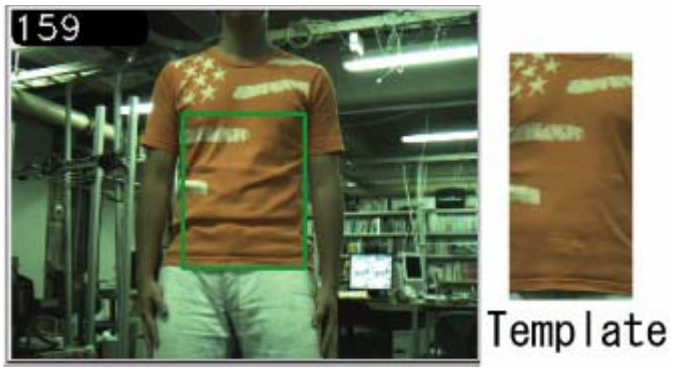


Fig.9 Person detecting image

recognize a specific person. The subject occurs that it cannot follow only a specific person. Therefore, when performing target person detection based on a distance sensor, another function is needed in order to recognize a target person.

Kwon et al. [5], Sidenbladh et al. [6] have proposed the method of detecting a target person based on a vision sensor and performing person following. Although robot can perform detection and pursuit of the target person, a target person is only one person at the environment under person following.

Moreover, view environment of the robot which performs image processing is also set as monotonous background environment be easy to detect a target person. It can be said that the method is a possible method only in the prepared specific environment. In actual environment, the information on a target person's color may change a lot due to lighting depending on a place. In order to realize to personal following in an everyday life, it is necessary to be the detection method which can respond robustly in such environment.

In this research, we use the parson detection system robust about intensity-of-illumination change which Ito et al. study. This is the method of using together distance information and the color information. The processing step is shown below.

First, distance information is acquired from stereo camera. An object and a background portion are separated. In this research, robot may always follow and keep in distance with a target person within the limits of 1 to 2 m. Therefore, the object which exists far away from 2m does not have necessity in image processing, and a robot does not need to look at it. Then, using the distance information by a stereo disparity, the robot considers that the 2m or more is a background, and does not use it for image processing. As a result, also in environment with a complicated field of view from the robot, the robot can remove a complicated background portion. Moreover, even if the color similar to an object exists in a background portion, the robot does not extract.

A HSV color system is use for the information on a color in this method. The robot creates the histogram which sets an axis of ordinate to S and a horizontal axis to H from a target person's template. However, the target person does not necessarily wear monochromatic dress. Then, the robot uses the NN method (curtate-distance method) based on the created H-S histogram, and performs clusterization of the histogram.

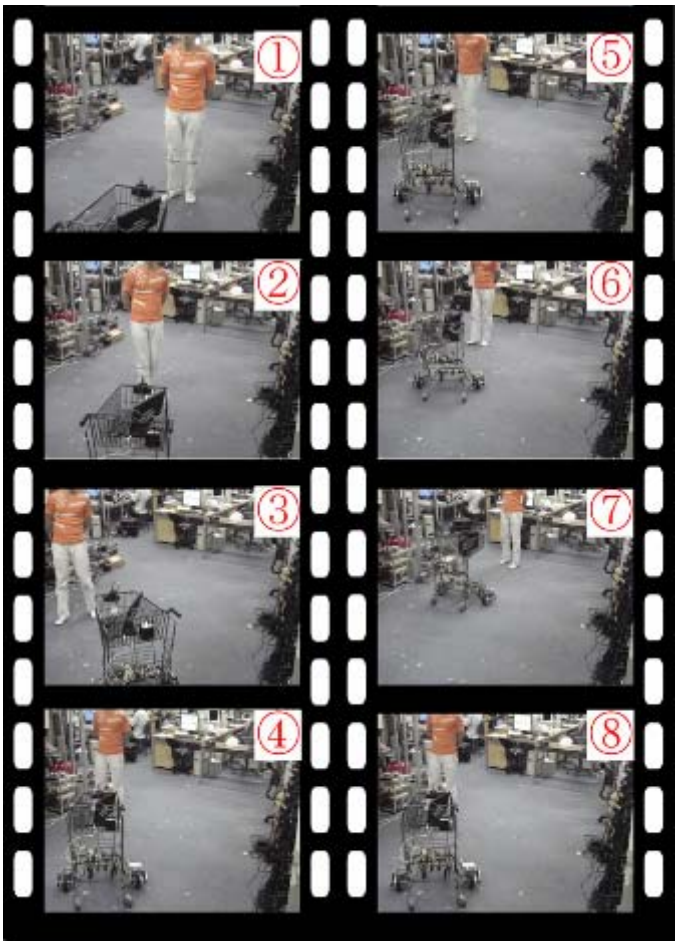


Fig.10 Examination of the person following

The robot performs weighting in the extraction color updated from a target person's detection position. Non-Gaussian distribution is use for weighting. The robot calculates non-gauss type distribution of each cluster of the cluster-ized H-S histogram. However, in this process, when two or more colors of a target person are in a picture, a problem occurs. Even if the robot deletes a background portion by a distance picture, the robot may extract the color which is not a target out of a picture. As a result, the robot may update the color of the area which is not a target person. Then, the whole cluster is weighted about the created non-Gaussian distribution.

In order to make following operation robust, renewal of a template is required. Then, the robot update a H-S histogram based on the information on the color of a target person's detection area. A processing picture when the robot performs the person detect processing is shown in Fig. 9. Fig. 9 shows a processing picture when we do a person following examination of the robot so that we may mention later. The picture on the right of Fig. 9 is a target person's template. The portion enclosed green is a detection area. The upper left number shows the distance [cm] from the robot to the target. From Fig.9, we check that the robot detects the target parson using the template.

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The running examination of the shopping cart robot is done. We do the examination of following while detecting a target under indoor environment. The sequential photographs of the experimental situation are shown in Fig. 10.

In Fig.10, the target person walks moving right and left. The shopping cart robot detects the target person and follows him. By this examination, we can check that the robot can follow a target person while carries baggage.

VI. CONCLUSION

We proposed a concept called robotizing daily items. As a concrete target, we set up development of the shopping cart robot which can follow a target person. The improved driving module was developed and tested the running performance. Using the design 2 of the driving module, the robot can run outdoors. The compatible control was used in which the change to manual control and automatic control is possible. We did the following examination indoors using the stereo camera, and check the robot of operation. Since the shopping cart robot can run also in the outdoors, we aim at realization of person following operation in outdoors.

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